

OSTRA: LEARNING MULTI-STAGE ROBOT MANIPULATION THROUGH OBJECT-STATE TRANSITION SUPERVISION

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ABSTRACT

Sparse supervision is a key challenge in multi-stage manipulation, with the lack of supervision over the past-to-future evolution of objects required by the policy being particularly critical. To address this issue, we introduce the Object-State Transition Map (**OST-Map**), a unified object-centric representation that organizes task-relevant objects as identity-consistent graphs and represents historical and future object states in a shared structured space. Building on OST-Map, we propose **OSTRA**, a diffusion-based Object-State Transition Reasoning Architecture that hierarchically predicts future object states, target TCP poses, and continuous action trajectories. OSTRA combines Transition-Guided Attention, which prioritizes historical object-state transitions during future-state prediction, with Staged Denoising, which progressively translates predicted object changes into robot sub-goals and executable actions. Experiments on three simulation benchmarks and seven diverse real-world tasks demonstrate consistent improvements over representative imitation learning baselines. Further ablations validate the contributions of OST-Map and the proposed architectural designs.

1 INTRODUCTION

Sparse supervision is a key challenge in imitation learning, particularly for multi-stage robot manipulation tasks (Zare et al., 2024; Mandlekar et al., 2020; Chen et al., 2023; Jain et al., 2025). Completing such tasks requires a policy not only to understand the environment at the current time step, but also to retain knowledge of the past—for example, whether prior subtasks succeeded and how key objects changed (Figure 1, left) (Kang & Kuo, 2025; Mark et al., 2026). Because an environment consists of objects, the policy must understand how these objects have changed over time. Therefore, the currently observed environment alone is insufficient for the policy to predict the most appropriate action. The policy must also be aware of object changes within the environment. We refer to this challenge as the object-state transition supervision gap.

Early imitation learning methods improve short-term action prediction through action chunking or limited observation windows, but do not explicitly capture long-horizon task-relevant state transitions (Zhao et al., 2023; Chi et al., 2025). Recent approaches incorporate past information through observation histories, latent memories, or task progress (Liang & Boularias, 2021; Shi et al., 2026; Mark et al., 2026). Others use future observations, goal states, or latent plans as additional supervision (Lynch et al., 2020; Mandlekar et al., 2020; Belkhale & Sadigh, 2023). However, these methods represent temporal information at observation, latent, or task level, providing limited supervision on individual-object evolution during execution. Our method bridges this gap by unifying both temporal directions at the object level. Specifically, it represents past object-state transitions as structured historical context and predicts future object states as auxiliary supervision, enabling the policy to reason about how task-relevant objects have changed and how they should evolve next.

To provide such supervision over past-to-future object evolution, we introduce the **Object-State Transition Map (OST-Map)**, an object-centric representation of observed and future states in a shared structured space. OST-Map avoids requiring policies to discover transition-relevant cues from visual histories using robot-centric action supervision alone. Specifically, OST-Map represents each task-relevant object as an identity-consistent graph of structural nodes. Each node encodes

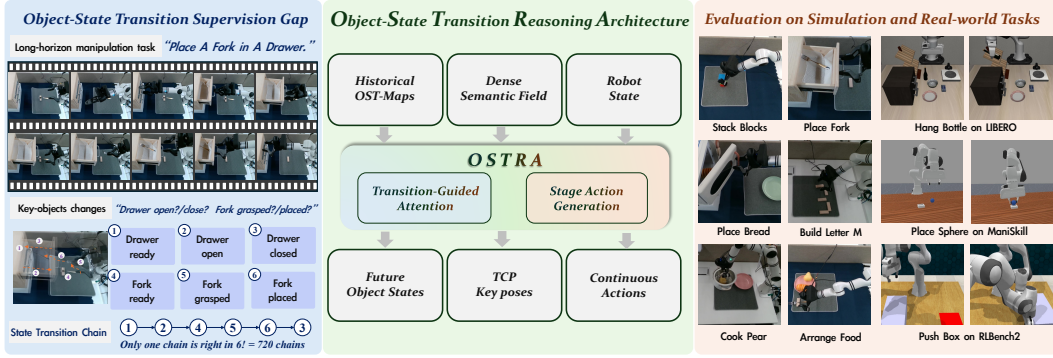


Figure 1: **Overview of OSTR.** Multi-stage manipulation requires resolving the object-state transition supervision gap: the policy must infer the correct transition chain from historical object changes. OSTR combines historical OST-Maps, a dense Semantic Field, and the robot state to predict future object states, TCP keyposes, and continuous actions, and is evaluated across simulation and real-world tasks.

its semantic identity, geometric parameters, and 6-DoF pose, while the graph edges capture structural and spatial relations among nodes and objects. By preserving object and node identities across time, OST-Map organizes observations into temporally ordered object-centric tokens that explicitly characterize how task-relevant objects evolve. It further represents future object states in the same structured space, allowing the transition from historical to future object states to be directly supervised. In this way, OST-Map turns otherwise implicit task progress and anticipated object changes into explicit intermediate targets between visual perception and robot action generation.

Building on OST-Map, we introduce OSTR, an **Object-State Transition Reasoning Architecture** for multi-stage manipulation (Figure 1, middle). OSTR is a diffusion-based framework that hierarchically predicts future object states, target TCP poses, and continuous action trajectories. It incorporates two key designs. First, Transition-Guided Attention attends to historical OST-Map tokens before incorporating dense Semantic Field tokens, encouraging the model to prioritize task-relevant object-state transitions when predicting future states. Second, Staged Denoising follows an object-to-TCP-to-action hierarchy that progressively translates predicted object-state changes into robot subgoals and executable actions. Together, these designs connect object-level transition reasoning with robot-level planning and continuous control, enabling the policy to reason from how objects have changed to how they should evolve next and which actions should realize those changes.

We evaluate OSTR on three simulation benchmarks and seven diverse real-world tasks. Across these settings, OSTR consistently improves over representative imitation learning baselines on multi-stage manipulation tasks. Further ablation studies validate the contributions of OST-Map, Transition-Guided Attention, and Staged Denoising, demonstrating the effectiveness of structured object-state transition supervision for connecting scene evolution with robot action generation.

Our contributions are: (1) To tackle the object-state transition supervision gap in multi-stage imitation learning, we introduce OST-Map, an object-centric representation that aligns historical object-state sequences with future object-state targets, enabling direct supervision of object-state transitions. (2) Building on OST-Map, we introduce OSTR, a diffusion-based architecture with two key designs. a) Transition-Guided Attention applies cross-attention to OST-Map tokens before Semantic Field tokens, encouraging the model to prioritize object-state transition information. b) Staged Denoising hierarchically predicts future object states, TCP poses, and action trajectories, progressively translating object-level state changes into robot subgoals and continuous control. (3) We conduct evaluations across three simulation benchmarks and seven diverse real-world tasks, demonstrating consistent performance gains over representative imitation learning baselines.

2 RELATED WORK

2.1 LONG-HORIZON VISUOMOTOR IMITATION LEARNING

Long-horizon manipulation requires coherent behavior over extended execution. Existing methods shorten the control horizon through action chunking, autoregressive prediction, or diffusion-based

action generation (Zhao et al., 2023; Chi et al., 2025; Haldar et al., 2024; Reuss et al., 2024). Other approaches introduce higher-level task structure through subgoals, skill composition, progress estimation, or planning (Shiari et al., 2018; Gupta et al., 2020; Chen et al., 2023; Kang & Kuo, 2025; Jain et al., 2025; Liu et al., 2026; Zhang et al., 2026b; Feng et al., 2025; Zhou et al., 2026). These methods organize actions or task stages over time, but generally do not supervise how task-relevant objects evolve from historical to target states.

2.2 TEMPORAL MODELING IN ROBOT POLICIES

Temporal context is commonly incorporated through observation histories, latent memories, recurrent features, or selectively retrieved past frames (Liang & Boularias, 2021; Fang et al., 2025; Shi et al., 2026; Mark et al., 2026). Recent policies further maintain recurrent or hierarchical memory to capture dependencies beyond the current observation (Ji et al., 2026; Xiao et al., 2026; Li et al., 2026).

A complementary direction uses future information as guidance. Existing methods condition policies on goal observations or latent plans (Lynch et al., 2020; Mandlekar et al., 2020), predict future visual states or joint video-action trajectories (Hu et al., 2025; Zhu et al., 2025), or learn predictive representations from future transitions and interaction targets (Zeng et al., 2024). Despite using historical and future context, these approaches represent time through holistic observations, latent memories, predicted frames, or task-level targets rather than identity-consistent state transitions of individual objects.

2.3 OBJECT-CENTRIC REPRESENTATIONS FOR MANIPULATION

Object-centric policies organize perception and action around manipulable entities. Existing approaches use object proposals or 3D object features as policy inputs (Zhu et al., 2023b;a), learn affordance-, flow-, or contact-based interaction representations (Belkhale & Sadigh, 2023; Xu et al., 2025; Zhang et al., 2026a; Lee & Fazeli, 2025), or represent manipulation through explicit object motion, including SE(3) pose trajectories and object pointflow (Hsu et al., 2025; Yang et al., 2025). Recent work also introduces structured object-level representations based on physical concepts or semantic operations (Wei et al., 2026; Tang et al., 2026).

Object-centric world models use entity-level representations for dynamics and future prediction (Ferraro et al., 2025; Jeong et al., 2025), while graph-based approaches preserve explicit objects and relations for manipulation planning (Zhu et al., 2026). These methods typically represent objects as perceptual units, interaction interfaces, motion variables, or future prediction targets. In contrast, OST-Map maintains identity-consistent structural states across observed and future task stages, making past-to-future object-state transitions explicit intermediate supervision for action generation.

3 METHOD

3.1 OVERVIEW

Problem Formulation. We consider multi-stage robot manipulation from RGB-D observations. At time t , the policy receives an observation window $\mathcal{O}_t = \{(I_\tau, D_\tau, s_\tau)\}_{\tau=t-T_o+1}^t$ and predicts an action chunk $A_t = \{a_{t+k}\}_{k=0}^{H_a-1}$. Rather than directly mapping the current visual observation to robot actions, OSTRA explicitly models how task-relevant objects have evolved throughout the observation history and which object states should be achieved next.

Object-State Transition Modeling. As shown in Figure 2, each RGB-D observation is converted into an OST-Map whose identity-consistent nodes encode object semantics, structure, affordance, and 6-DoF pose. Tracking yields aligned historical maps that expose object-state transitions and supervise future states in the same structured space, while a dense Semantic Field preserves local scene evidence.

Transition-Guided Action Generation. Transition-Guided Attention first grounds prediction queries in historical OST-Map tokens and then incorporates dense Semantic Field tokens. Staged Action Generation follows the hierarchy *object state* $\rightarrow$ *TCP subgoal* $\rightarrow$ *action*: future OST-Map

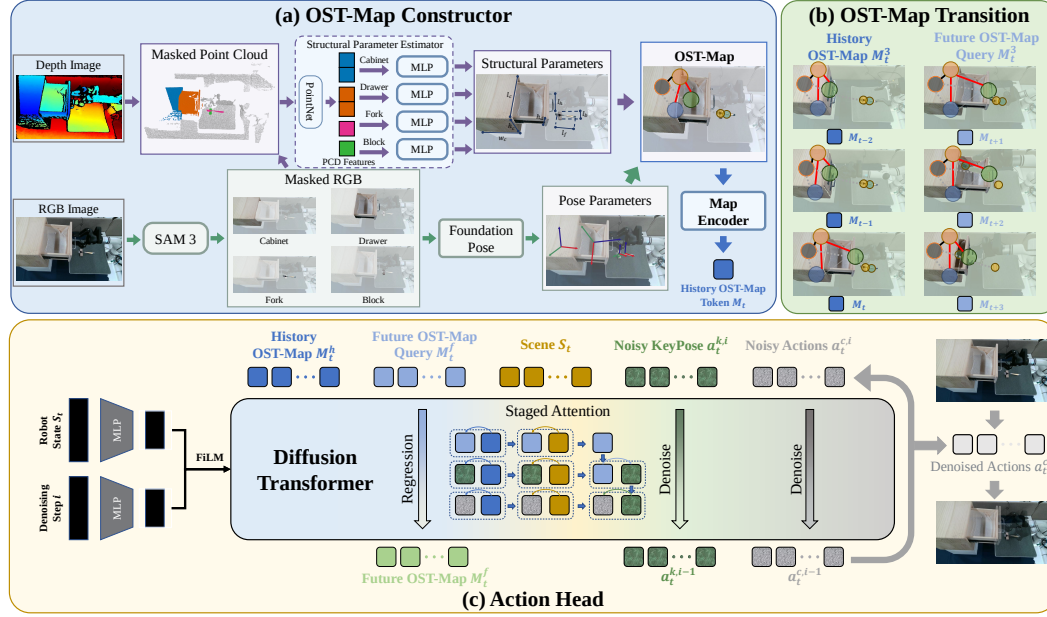


Figure 2: **Overview of OSTR.** (a) **OST-Map Constructor:** RGB-D observations are converted into identity-consistent object-centric maps by estimating task-relevant object masks, structural parameters, and 6-DoF poses. (b) **Object-State Transition Modeling:** temporally aligned historical OST-Maps provide explicit object-state transition context and are used to query future object states. (c) **Transition-Guided Action Generation:** historical OST-Map tokens, future object-state queries, dense Semantic Field tokens, robot states, and the diffusion timestep are processed by OSTR, which progressively predicts future object states, TCP keyposes, and continuous actions following the hierarchy *object state* $\rightarrow$ *TCP subgoal* $\rightarrow$ *action*.

keyframes condition TCP-keyframe denoising, whose representation then conditions continuous-action generation.

3.2 OST-MAP: OBJECT-STATE TRANSITION MAP

As illustrated in Figure 3, OST-Map composes parameterized primitives into object parts and identity-consistent structural nodes. Persistent node identities preserve fine-grained structure and enable temporal tracking.

For a task with N structural nodes, the OST-Map at time t is defined as an attributed graph $\mathcal{M}_t = (\mathcal{V}_t, \mathcal{E}_t)$. Each node $v_{t,i} \in \mathcal{V}_t$ is represented as

$$v_{t,i} = (c_i, \mathcal{P}_i, \mathcal{A}_i, p_{t,i}, q_{t,i}), \quad (1)$$

where c_i denotes its semantic identity, $\mathcal{P}_i$ and $\mathcal{A}_i$ denote its structural and affordance point sets, and $(p_{t,i}, q_{t,i}) \in \mathbb{R}^3 \times \mathbb{S}^3$ denotes its 6-DoF pose. Semantic identity and local structural attributes remain fixed within an episode, while pose evolves with the object state.

Each edge $e_{t,ij} \in \mathcal{E}_t$ stores its relation type, parameters, reference anchors, and relative pose; free-relation edges connect pairs without task-specific relations. The sequence $\mathcal{M}_{t-T_o+1:t}$ captures object-state evolution in an identity-aligned space.

3.3 OSTR: OBJECT-STATE TRANSITION REASONING ARCHITECTURE

OST-Map Construction and Transition Targets. As shown in Figure 2(a), OSTR first constructs an OST-Map from each RGB-D observation. SAM 3 (Carion et al., 2026) segments prompted task objects, FoundationPose (Wen et al., 2024) registers and tracks their 6-DoF poses, and PointNet (Charles et al., 2017) with object-specific MLP heads estimates structural parameters from masked point clouds, producing a temporally aligned map sequence.

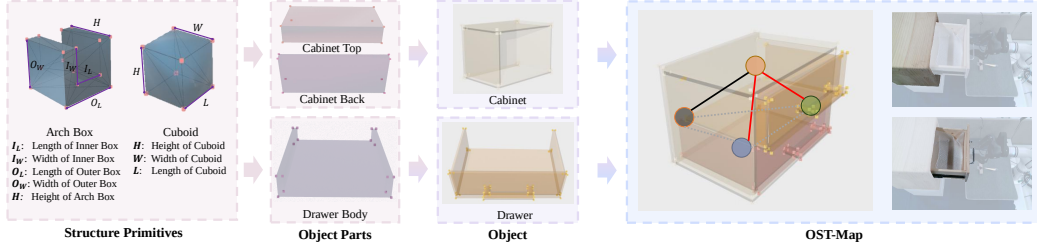


Figure 3: **Representation of OST-Map.** Parameterized structure primitives are composed into object parts and further assembled into task-relevant objects. Each object is represented by identity-consistent structural nodes and their relations, while node poses evolve with manipulation. The resulting OST-Map provides a compact object-centric representation of scene structure and object-state transitions.

Historical OST-Maps encode observed transitions. During training, the structural-node positions at the next H_k task keyframes define future-state targets,

$$Y_t^M = \{p_{t,h,i}^*\}_{h=1, i=1}^{H_k, N}. \quad (2)$$

paired with TCP keyframes Y_t^E and action chunk A_t . Future observations provide training-only supervision, while the shared OST-Map space supplies identity-aligned transition targets.

Multimodal Context Encoding. OSTRa combines a sparse OST-Map with a dense Semantic Field. Depth points $X_t = \{x_{t,j}\}_{j=1}^P$ are paired with image-aligned pretrained DINO features $d_{t,j}$. PointNet++ encodes $[x_{t,j}; d_{t,j}]$ into dense tokens F_t^S for cross-attention and downsampled tokens $\tilde{F}_t^S$ for compact planning context.

For each historical map, node embeddings combine PointNet features of $\mathcal{P}_i$ and $\mathcal{A}_i$ with semantic, pose, temporal, and identity embeddings; edge embeddings encode relation attributes and relative pose. A graph transformer produces coordinate-aware map tokens F_t^M . Map and scene tokens retain 3D coordinates for relative spatial attention; robot state and diffusion timestep condition prediction blocks through adaptive normalization.

Transition-Guided Attention. A central challenge is the large imbalance between the sparse OST-Map tokens and the dense Semantic Field tokens. Directly mixing them in a single attention context can weaken the structured transition cues carried by the OST-Map. OSTRa therefore introduces *Transition-Guided Attention*, in which each prediction query first attends to historical OST-Map tokens and subsequently to the dense Semantic Field:

$$\tilde{Z} = \text{CA}(Z, F^M), \quad Z^+ = \text{CA}(\tilde{Z}, F^S), \quad (3)$$

where CA denotes cross-attention. Cross-attention uses relative 3D rotary encodings computed from query and context coordinates.

For future object-state prediction, the queries are constructed by repeating the latest observed structural-node coordinates over the next H_k keyframes and augmenting them with learned keyframe-time and node-identity embeddings. TCP and action queries come from their noisy diffusion variables. The map-first order preserves transition grounding before introducing denser visual context.

Staged Action Generation. As illustrated in Figure 2(c), OSTRa translates object-state transitions into executable robot motion through the hierarchy *object state* $\rightarrow$ *TCP subgoal* $\rightarrow$ *action*. At each diffusion step, future-object queries yield an object plan P^M and predicted states $\hat{Y}^M$. Noisy TCP queries then combine ordered map-to-scene attention with $\text{sg}(P^M)$ to produce a TCP plan P^E and TCP-noise estimate. Finally, noisy action queries condition on $\text{sg}(P^E)$ to predict action noise and gripper openness. TCP keyframes and actions are progressively denoised from Gaussian noise, whereas future object states are re-predicted at every step. Position and rotation use separate schedules. Algorithm 1 summarizes the procedure.

Training Objective and Inference. OSTRa is trained with direct supervision at all three levels of the hierarchy. The objective combines future object-state prediction, TCP diffusion, action diffusion,

Algorithm 1 Staged Action Generation in OSTR

Require: Map context F^M , scene context F^S , robot state s_t
Ensure: Action chunk $\hat{A}_t$

- 1: Construct future object queries Z^M
- 2: Initialize $Y^{E,N}, A^N \sim \mathcal{N}(0, I)$
- 3: **for** $n = N, \dots, 1$ **do**
- 4: $P^M, \hat{Y}_n^M \leftarrow \text{OBJECTSTAGE}(Z^M, F^M, F^S, s_t, n)$
- 5: $P^E, \hat{\epsilon}_n^E \leftarrow \text{TCPSTAGE}(Y^{E,n}, F^M, F^S, \text{sg}(P^M), s_t, n)$
- 6: $P^A, \hat{\epsilon}_n^A, \hat{G}_n \leftarrow \text{ACTIONSTAGE}(A^n, F^M, F^S, \text{sg}(P^E), s_t, n)$
- 7: $Y^{E,n-1} \leftarrow \text{DENOISE}(Y^{E,n}, \hat{\epsilon}_n^E, n)$
- 8: $A^{n-1} \leftarrow \text{DENOISE}(A^n, \hat{\epsilon}_n^A, n)$
- 9: **end for**
- 10: **return** A^0 with predicted gripper states

and gripper supervision:

$$\begin{aligned} \mathcal{L} = & \lambda_M \|\hat{Y}^M - Y^M\|_1 + \lambda_E \|\hat{\epsilon}_E - \epsilon_E\|_1 \\ & + \lambda_A \|\hat{\epsilon}_A - \epsilon_A\|_1 + \lambda_G \|\hat{G} - G\|_1. \end{aligned} \quad (4)$$

The terms supervise object evolution, TCP subgoals, continuous control, and gripper state. Stop-gradient requires each stage to satisfy its own target.

At inference, map and field contexts are encoded once per replanning step. Each reverse step executes Algorithm 1; the policy then executes the first predicted actions and replans from the updated history.

4 EXPERIMENTS

Our experiments ask whether OSTR improves simulation (Section 4.1) and real-world manipulation (Section 4.2); how foundation-model errors and cost affect it (Sections 4.3 and 4.4); and which designs and hyperparameters matter (Section 4.5).

4.1 SIMULATION EXPERIMENTS

Benchmarks and Experimental Setup. We benchmark OSTR on three manipulation suites, with representative task rollouts shown in Figure 4. **RLBench2**, based on RLBench (James et al., 2020), contains seven multi-stage tasks requiring long-horizon object-state reasoning. **ManiSkill** (Gu et al., 2023) contains six contact-rich insertion, placement, tool-use, and stacking tasks. **LIBERO** (Liu et al., 2023) contains four ten-task suites—Spatial, Object, Goal, and Long—for diverse language-conditioned manipulation.

We use official datasets and protocols and report task success rate. Detailed settings are provided in Appendix A.

Baselines. We compare OSTR with benchmark-specific methods spanning action generation, history and memory, future prediction, and object-centric perception and motion (Zhao et al., 2023; Chi et al., 2025; Ze et al., 2024; Fang et al., 2025; Yang et al., 2025; Zhu et al., 2023a; Hsu et al., 2025). Detailed descriptions of all baselines are provided in Appendix A.1.

Simulation results. As summarized in Table 1, OSTR achieves 89.8%, 74.5%, and 97.4% on RLBench2, ManiSkill, and LIBERO, improving over their strongest reported baselines by 5.3, 17.7, and 0.7 points. Appendix A.2 provides full per-task results.

4.2 REAL-WORLD EXPERIMENTS

We evaluate OSTR on a 7-DoF Realman 75-BF arm with a Pika parallel gripper. An Intel RealSense D415 captures RGB-D observations, and Open3D (Zhou et al., 2018) reconstructs point

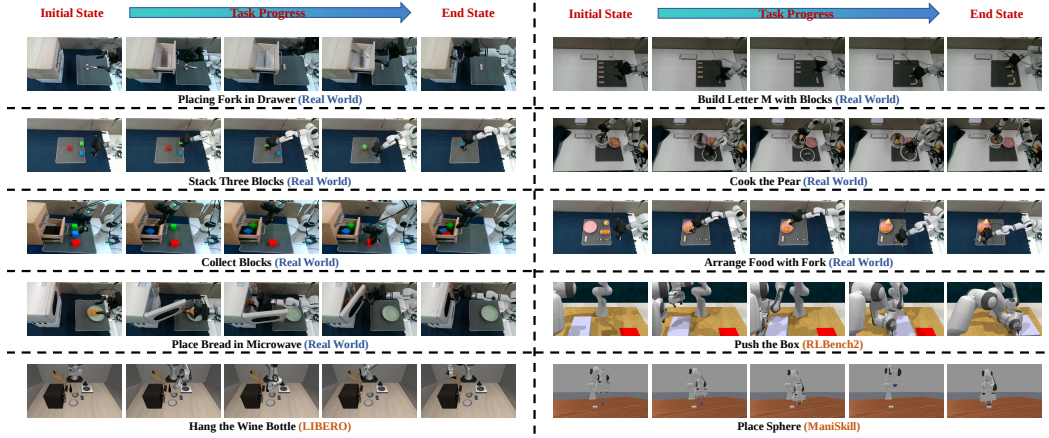


Figure 4: Task demonstrations from initial state to task completion. The seven real-world tasks are *Place Fork in Drawer*, *Stack Three Blocks*, *Collect Blocks*, *Place Bread in Microwave*, *Build Letter M with Blocks*, *Cook the Pear*, and *Arrange Food with Fork*. The figure also shows representative tasks from the three simulation benchmarks: *Push the Box* from RLBench2 (James et al., 2020), *Place Sphere* from ManiSkill (Gu et al., 2023), and *Hang the Wine Bottle* from LIBERO (Liu et al., 2023).

Table 1: **Summary of the main simulation results.** We report average task success rate (%) on each benchmark. Full per-task results are provided in Appendix A.2. Best results are in **bold**, and second-best results are underlined.

RLBench2 (James et al., 2020) (7 tasks)		ManiSkill (Gu et al., 2023) (6 tasks)		LIBERO (Liu et al., 2023) (40 tasks)	
Method	Mean S.R. (%)	Method	Mean S.R. (%)	Method	Mean S.R. (%)
ACT (Zhao et al., 2023)	15.5	ACT (Zhao et al., 2023)	21.3	ACT (Zhao et al., 2023)	67.7
DP3 (Ze et al., 2024)	26.0	Diffusion Policy (Chi et al., 2025)	26.8	GROOT (Zhu et al., 2023a)	66.2
3D Diffuser Actor (Ke et al., 2025)	64.7	DP3 (Ze et al., 2024)	53.8	Diffusion Policy (Chi et al., 2025)	72.4
PerAct ² (Shridhar et al., 2023)	40.0	3D Diffuser Actor (Ke et al., 2025)	51.5	BAKU (Haldar et al., 2024)	93.0
PPI (Yang et al., 2025)	80.8	PPI (Yang et al., 2025)	<u>56.8</u>	MDT (Reuss et al., 2024)	76.1
Diffusion Policy (Chi et al., 2025)	42.6	BAKU (Haldar et al., 2024)	43.0	DP3 (Ze et al., 2024)	73.0
SAM2Act+ (Fang et al., 2025)	<u>84.5</u>	FlowPolicy (Zhang et al., 2025)	39.5	UWM (Zhu et al., 2025)	91.8
GROOT (Zhu et al., 2023a)	45.2	SPOT (Hsu et al., 2025)	47.5	MemoryVLA (Shi et al., 2026)	<u>96.7</u>
SPOT (Hsu et al., 2025)	82.3	Im2Flow2Act (Xu et al., 2025)	45.7	BPP (Mark et al., 2026)	90.9
OSTRA (Ours)	89.8	OSTRA (Ours)	74.5	OSTRA (Ours)	97.4

clouds. The map constructor uses predicted masks, poses, and structural parameters, evaluating reasoning under errors accumulated by the OST-Map perception pipeline.

Task suite. We consider seven tasks spanning articulated interaction, long-horizon rearrangement, thin or reflective objects, and global spatial constraints. They require opening and closing containers, sequential stacking, cluttered placement, partial-occlusion reasoning, pattern construction, collection, and shared-workspace manipulation. As visualized in Figure 4, they cover articulated state, support, containment, occupancy, and multi-object layout transitions.

Real-world results. Across 20 trials per task (Table 2), OSTRA averages 84.3%, exceeds SAM2Act+ (59.3%) by 25.0 points, and ranks first on all tasks. It is more reliable on drawer and microwave tasks and better preserves relations in stacking and shape construction than Diffusion Policy (Chi et al., 2025) and DP3 (Ze et al., 2024). Its largest gain is on *Build Letter M with Blocks* (65% versus 15%), supporting object-state transition modeling under noisy, partially occluded observations.

4.3 SYSTEM ERROR BREAKDOWN

We use ground-truth replacements to identify which OST-Map Constructor errors most affect task success.

Setup. We replace outputs of SAM 3 extraction, FoundationPose estimation and tracking, and PointNet-MLP structural estimation with ground-truth masks, poses, or sizes while fixing the policy and evaluation. We test all eight combinations: none, each single or pairwise replacement, and all GT. Ground truth is used only for diagnosis.

Table 2: **Real-world manipulation results.** We report full-task success rate (%) over 20 trials per task and the average across seven tasks. Best results are shown in **bold**, and second-best results are underlined.

Method	Avg. $\uparrow$	Place Fork in Drawer	Stack Three Blocks	Arrange Food with Fork	Place Bread in Microwave	Build Letter M with Blocks	Collect Blocks	Cook the Pear
ACT (Zhao et al., 2023)	19.3	15	10	35	20	0	30	25
Diffusion Policy (Chi et al., 2025)	20.0	10	5	40	20	0	35	30
DP3 (Ze et al., 2024)	40.7	40	20	65	45	0	60	55
PPI (Yang et al., 2025)	50.7	50	30	75	55	10	70	65
SAM2Act+ (Fang et al., 2025)	<u>59.3</u>	<u>55</u>	<u>40</u>	<u>85</u>	<u>65</u>	<u>15</u>	<u>80</u>	<u>75</u>
MemoryVLA (Shi et al., 2026)	54.3	50	35	80	60	10	75	70
MPI (Zeng et al., 2024)	44.3	45	25	70	50	0	65	55
SPOT (Hsu et al., 2025)	52.1	45	35	80	55	10	75	65
Im2Flow2Act (Xu et al., 2025)	42.9	40	25	70	45	5	60	55
OSTRA (Ours)	84.3	85	80	95	85	65	90	90

Analysis. Table 3 reports average success relative to Full OSTRA. Single and joint replacements reveal individual and interacting errors. The Semantic Field encoder and policy remain unchanged, so all-GT residual failures are not solely due to map construction.

4.4 LATENCY

Setup. We profile batch-one inference on an NVIDIA GeForce RTX 4090. The first frame segments masks, registers poses, estimates structure, and initializes OST-Map; later frames track objects and update the map. Both include encoding and policy inference.

Results. Table 4 reports 2835.325 ms for initialization and 226.292 ms (4.419 FPS) thereafter, a $12.5\times$ speedup. Segmentation and pose estimation dominate; map construction, encoding, and policy inference add little overhead.

4.5 ABLATION STUDIES

We ablate OST-Map temporal roles and attributes, map–visual attention order, staged denoising and stop-gradient, and temporal and loss hyperparameters. All variants use the same RL Bench2 data, capacity, budget, and evaluation seeds.

OST-Map. Table 5 compares no map or future loss (Ex1), history only (Ex2), future supervision from the latest map (Ex3), and the full model (Ex11). Ex4–Ex11 progressively add geometry, semantics, affordances, and edge attributes to pose-only nodes under the same graph encoder and node correspondence. Full OST-Map reaches 89.8%, 20.4 points above no map; history-only and future-only reach 84.5% and 85.8%. Their combination and the Ex4–Ex11 trend show that both temporal roles and attribute groups help.

Transition-Guided Attention. Ex1–Ex4 in Table 6 test map-only, visual-only, joint, and visual-first attention against map-first (Ex12), with the full hierarchy and stop-gradient boundaries. Map-first reaches 89.8%, exceeding these variants by 13.3, 11.6, 4.4, and 2.7 points, respectively.

Staged Denoising. With map-first attention fixed, Ex5–Ex8 test action-only, object-to-action, TCP-to-action, and flat joint prediction; Ex9–Ex12 test object/TCP stop-gradient combinations. Removed stages omit their heads and losses, while the flat variant retains all targets without hierarchical conditioning. The full hierarchy exceeds action-only by 14.0 points and flat prediction by 8.3; both stop-gradient boundaries add 3.9 points over neither.

Hyperparameter sensitivity. Figure 5 varies four temporal counts—past keyframes H_p , observations T_o , future keyframes H_k , and actions H_a —and three loss ratios: λ_M/λ_A , λ_E/λ_A , and λ_G/λ_A . Each curve varies one parameter around the 89.8% full-model reference. Performance is stable nearby but falls at short or long horizons; H_a , H_p , and object-state loss are most sensitive, while gripper loss is stable.

Table 3: System error breakdown for OSTR. We use ground-truth replacements to identify the dominant bottleneck in OST-Map construction. Ground-truth quantities are used only for analysis and are not part of the inference interface. Δ reports the improvement over Full OSTR.

Setting	GT Extraction	GT Object Pose	GT Structural Size	Avg. Success	Δ
Full OSTR				89.8	0.0
+ GT Extraction	✓			90.3	+0.5
+ GT Object Pose		✓		91.9	+2.1
+ GT Structural Size			✓	93.4	+3.6
+ GT Extraction + GT Object Pose	✓	✓		92.1	+2.3
+ GT Extraction + GT Structural Size	✓		✓	94.2	+4.4
+ GT Object Pose + GT Structural Size		✓	✓	95.7	+5.9
+ GT Extraction + GT Object Pose + GT Structural Size	✓	✓	✓	96.4	+6.6

Table 4: **Inference latency (ms) and throughput.** We report mean component and end-to-end latency for the first frame and subsequent frames at batch size one. The first frame initializes OST-Map; subsequent frames track objects and update the map.

Stage	Mask Seg. / Track. ↓	Pose Reg. / Track. ↓	Structure & Map ↓	Encoders & Policy ↓	Total (ms) ↓	FPS ↑
First Frame (Initialization)	1412.350	1386.425	21.340	15.210	2835.325	0.353
Later Frames (Tracking)	94.832	96.155	20.125	15.180	226.292	4.419

Table 5: **OST-Map ablations on RLbench2.** We vary temporal supervision and map attributes with a fixed graph encoder. Mean is average success (%); Ex11 is the full model.

	Temporal Hist. Fut.	Node Geo Pose Sem Aff	Edge Type Param Anchor RelPose	Mean
Ex1	- -	- - - -	- - - -	69.4
Ex2	✓ -	✓ ✓ ✓ ✓	✓ ✓ - ✓	84.5
Ex3	- ✓	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	85.8
Ex4	✓ ✓	- ✓ - -	- - - -	75.1
Ex5	✓ ✓	✓ ✓ - -	- - - -	78.8
Ex6	✓ ✓	✓ ✓ ✓ -	- - - -	82.5
Ex7	✓ ✓	✓ ✓ ✓ ✓	- - - -	85.2
Ex8	✓ ✓	✓ ✓ ✓ ✓	✓ - - -	86.6
Ex9	✓ ✓	✓ ✓ ✓ ✓	✓ ✓ - -	87.8
Ex10	✓ ✓	✓ ✓ ✓ ✓	✓ ✓ ✓ -	88.9
Ex11	✓ ✓	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	89.8

Table 6: **OSTRA ablations on RLbench2.** We vary attention order, prediction stages, and stop-gradient boundaries. Mean is average success (%); Ex12 is the full model.

	Map	Attention Vis. $M \rightarrow V$ $V \rightarrow M$	Denoising Obj. TCP Hier.	Stop-Grad. Obj. TCP	Mean
Ex1	✓ - - -	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	76.5
Ex2	- ✓ - -	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	78.2
Ex3	✓ ✓ - -	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	85.4
Ex4	✓ ✓ - -	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	87.1
Ex5	✓ ✓ ✓ -	- - - -	- - - -	- - - -	75.8
Ex6	✓ ✓ ✓ -	- - - -	✓ - ✓ -	✓ - ✓ -	83.4
Ex7	✓ ✓ ✓ -	- - - -	✓ - ✓ -	✓ - ✓ -	84.6
Ex8	✓ ✓ ✓ -	- - - -	✓ ✓ ✓ -	- - - -	81.5
Ex9	✓ ✓ ✓ -	- - - -	✓ ✓ ✓ ✓	- - - -	85.9
Ex10	✓ ✓ ✓ -	- - - -	✓ ✓ ✓ ✓	✓ - ✓ -	87.4
Ex11	✓ ✓ ✓ -	- - - -	✓ ✓ ✓ ✓	✓ - ✓ -	88.3
Ex12	✓ ✓ ✓ -	- - - -	✓ ✓ ✓ ✓	✓ ✓ ✓ ✓	89.8

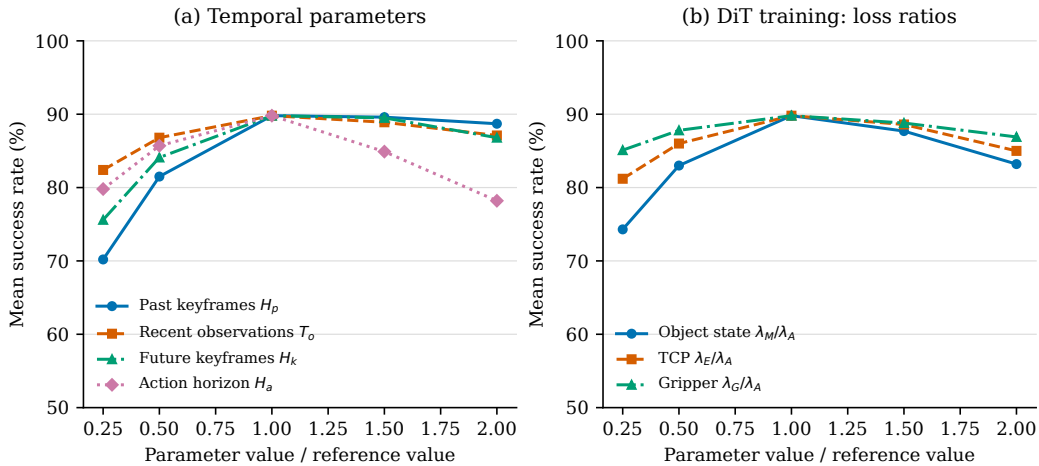


Figure 5: Hyperparameter sensitivity on RLbench2. (a) Past keyframes H_p , consecutive observations T_o , future keyframes H_k , and predicted actions H_a . (b) Object-state, TCP, and gripper loss weights relative to the action loss. Each curve varies one parameter with all others fixed at their reference values. The reference values are $H_p = T_o = H_k = 4$, $H_a = 16$, and $\lambda_M/\lambda_A = \lambda_E/\lambda_A = \lambda_G/\lambda_A = 1$.

5 CONCLUSION

We presented OSTR, a framework for multi-stage manipulation that represents historical and future object states in a shared Object-State Transition Map. Transition-Guided Attention and Staged Action Generation progressively translate object-state changes into TCP subgoals and executable actions. Across RL-Bench2, ManiSkill, LIBERO, and seven real-world tasks, OSTR consistently outperforms representative imitation-learning baselines, while the ablations validate the contributions of OST-Map and the proposed architectural designs. Future work will improve map construction under perception errors and extend object-state reasoning to less structured, open-world manipulation.

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A APPENDIX

A.1 BASELINE DETAILS

We compare OSTR with benchmark-specific baselines from four categories.

Action-generation policies. ACT (Zhao et al., 2023) uses a conditional variational transformer to predict temporally coherent action chunks. Diffusion Policy (DP) (Chi et al., 2025) generates continuous action sequences through conditional denoising. DP3 (Ze et al., 2024) conditions action diffusion on compact 3D point-cloud representations. 3D Diffuser Actor (Ke et al., 2025) performs policy diffusion over actions using language-conditioned 3D scene features. PerAct² (Shridhar et al., 2023) predicts discretized 6-DoF actions from language-conditioned voxel observations with a Perceiver-based policy. BAKU (Haldar et al., 2024) is an efficient transformer policy designed for multi-task imitation learning. FlowPolicy (Zhang et al., 2025) learns a fast 3D visuomotor policy through consistency flow matching. MDT (Reuss et al., 2024) is a diffusion transformer conditioned on multiple forms of task-goal input.

History and memory policies. SAM2Act+ (Fang et al., 2025) augments visual foundation-model features with a memory architecture for manipulation. MemoryVLA (Shi et al., 2026) equips a vision-language-action policy with perceptual and cognitive memory. BPP (Mark et al., 2026) handles long observation histories by selecting and attending to key past frames.

Future-prediction methods. PPI (Yang et al., 2025) represents manipulation through predicted gripper poses and object point flows. UWM (Zhu et al., 2025) jointly models future video and actions with coupled diffusion objectives. MPI (Zeng et al., 2024) pretrains manipulation representations by predicting interactions and therefore requires a separately specified downstream action head.

Object-centric perception and motion methods. GROOT (Zhu et al., 2023a) learns generalizable manipulation policies from object-centric 3D representations. SPOT (Hsu et al., 2025) generates object-centric SE(3) pose trajectories with diffusion. Im2Flow2Act (Xu et al., 2025) uses predicted object flow as a cross-domain interface from videos to robot actions.

A.2 FULL SIMULATION RESULTS

This section reports the full per-task results summarized in Table 1.

Table 7: **Quantitative results on RL Bench2.** We report task success rate (%) as mean $\pm$ standard deviation over three random seeds, with 100 evaluation rollouts per task for each seed. Best results are in **bold**, and second-best results are underlined.

Method	Avg. $\uparrow$	Box	Ball	Drawer	Laptop	Dustpan	Tray	Handover
ACT (Zhao et al., 2023)	15.5	67.0 $\pm$ 7.0	38.3 $\pm$ 10.0	1.7 $\pm$ 2.1	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	1.3 $\pm$ 1.5	0.0 $\pm$ 0.0
DP3 (Ze et al., 2024)	26.0	39.3 $\pm$ 3.1	27.0 $\pm$ 6.6	0.0 $\pm$ 0.0	6.0 $\pm$ 2.6	98.7 $\pm$ 0.6	6.3 $\pm$ 0.6	4.7 $\pm$ 1.5
3D Diffuser Actor (Ke et al., 2025)	64.7	54.7 $\pm$ 3.8	87.3 $\pm$ 1.9	52.7 $\pm$ 14.1	40.7 $\pm$ 6.1	96.7 $\pm$ 2.6	76.0 $\pm$ 4.3	44.7 $\pm$ 3.3
PerAct ² (Shridhar et al., 2023)	40.0	62.0 $\pm$ 26.2	50.0 $\pm$ 8.7	49.7 $\pm$ 16.6	36.7 $\pm$ 5.7	2.0 $\pm$ 3.5	60.0 $\pm$ 6.2	19.7 $\pm$ 6.0
PPI (Yang et al., 2025)	80.8	96.7 $\pm$ 1.5	89.3 $\pm$ 1.5	79.7 $\pm$ 3.8	46.3 $\pm$ 1.2	98.7 $\pm$ 1.5	92.0 $\pm$ 1.0	62.7 $\pm$ 2.5
DP (Chi et al., 2025)	42.6	61.0 $\pm$ 2.0	59.7 $\pm$ 3.5	38.5 $\pm$ 4.0	10.3 $\pm$ 3.1	69.7 $\pm$ 2.5	67.0 $\pm$ 3.2	2.0 $\pm$ 1.0
SAM2Act+ (Fang et al., 2025)	<u>84.5</u>	<u>95.2</u> $\pm$ 1.2	88.0 $\pm$ 2.0	<u>81.5</u> $\pm$ 3.0	52.0 $\pm$ 4.5	97.0 $\pm$ 1.1	<u>89.5</u> $\pm$ 1.8	<u>68.5</u> $\pm$ 3.2
GROOT (Zhu et al., 2023a)	45.2	65.0 $\pm$ 5.1	58.3 $\pm$ 4.2	35.0 $\pm$ 6.0	22.1 $\pm$ 3.8	45.0 $\pm$ 5.0	50.0 $\pm$ 4.0	41.2 $\pm$ 3.5
SPOT (Hsu et al., 2025)	82.3	94.0 $\pm$ 1.8	<u>86.5</u> $\pm$ 2.2	78.0 $\pm$ 4.1	<u>48.5</u> $\pm$ 3.9	96.5 $\pm$ 1.5	88.0 $\pm$ 2.1	64.8 $\pm$ 3.0
OSTRA (Ours)	89.8	98.3 $\pm$ 0.6	93.0 $\pm$ 1.5	87.0 $\pm$ 7.5	82.7 $\pm$ 9.6	<u>97.7</u> $\pm$ 1.0	92.0 $\pm$ 0.6	77.7 $\pm$ 3.8

Table 8: **Quantitative results on ManiSkill.** We report task success rate (%) and the average across six tasks. A dagger marks a published reference score, not a matched-data evaluation. “-” denotes an unavailable result or a pending evaluation. Best results are highlighted in **bold**.

Method	Avg. Success $\uparrow$	Peg Insertion Side	Place Sphere	Plug Charger	Pull Cube with Tool	Stack Cube	Stack Pyramid
ACT (Zhao et al., 2023)	21.3	0	2	28	38	49	11
Diffusion Policy (Chi et al., 2025)	26.8	0	4	36	47	61	13
DP3 (Ze et al., 2024)	53.8	55	62	58	48	72	28
3D Diffuser Actor (Ke et al., 2025)	51.5	48	60	55	46	71	29
PPI (Yang et al., 2025)	56.8	57	65	60	52	75	32
BAKU (Haldar et al., 2024)	43.0	25	44	48	55	66	20
FlowPolicy (Zhang et al., 2025)	39.5	14	32	45	53	68	25
SPOT (Hsu et al., 2025)	47.5	33	51	51	57	71	22
Im2Flow2Act (Xu et al., 2025)	45.7	29	47	52	56	69	21
OSTRA (Ours)	74.5	70	84	75	79	89	50

Table 9: **Quantitative results on LIBERO.** We report task success rate (%), with 50 evaluation trials per task. Best results are in **bold**, and second-best results are underlined.

Method	LIBERO-Spatial	LIBERO-Object	LIBERO-Goal	LIBERO-Long	Avg. Success $\uparrow$
ACT (Zhao et al., 2023)	82.0	78.8	66.1	44.0	67.7
GROOT (Zhu et al., 2023a)	79.1	81.5	63.2	41.0	66.2
Diffusion Policy (Chi et al., 2025)	78.3	92.5	68.3	50.5	72.4
BAKU (Haldar et al., 2024)	94.0	95.0	96.0	87.0	93.0
MDT (Reuss et al., 2024)	78.5	87.5	73.5	64.8	76.1
DP3 (Ze et al., 2024)	86.5	84.0	71.5	50.0	73.0
UWM (Zhu et al., 2025)	93.5	93.8	94.2	85.5	91.8
MemoryVLA (Shi et al., 2026)	<u>98.4</u>	<u>98.4</u>	<u>96.4</u>	<u>93.4</u>	<u>96.7</u>
BPP (Mark et al., 2026)	92.0	94.5	93.0	84.0	90.9
OSTRA (Ours)	98.5	98.7	97.3	95.0	97.4